

3x fewer servers, allowing Meta to consolidate a 60-server Java cluster down to just 20 Velox nodes.

The distributed batch benchmark:

Apache Spark + Gluten + Velox on AWS EKS: To evaluate how Velox scales for heavy-duty batch and ETL processing, Amazon Web Services (AWS) ran a TPC-DS 3TB benchmark on Amazon EKS using Graviton4 nodes (r8gd.12xlarge). This compared native Spark SQL against Spark accelerated by Apache Gluten and Velox.

- **Overall speedup:** The Gluten + Velox stack was 39% faster overall, dropping total suite runtime from 3,650.56 seconds to 2,239.93 seconds (a 1.63x overall speedup). Out of 103 queries, 81 queries saw significant improvements.
- **Deep dive on Query 93 (4.36x speedup):** In native Spark, Query 93 took 207.9 seconds, which Velox slashed to 47.7 seconds. By replacing Spark's traditional SortMergeJoin with Velox's ShuffledHashJoinExecTransformer and eliminating pre-join Sorts:
 - The hot 200-task join stage ran ~12x faster (dropping from 1m 16s to 6.5s). Peak per-task memory fell from 3.3GB (on-heap) to just 80MB (off-heap). JVM Garbage Collection time collapsed from 3.6 seconds to a mere 35 milliseconds.
 - Serialised shuffle data footprint was reduced by 24% (compressing from 214.1GB to 162.2GB), drastically reducing

disk and network pressure.

- **The whole-stage codegen nuance:** The benchmark also revealed interesting structural trade-offs. A small subset of queries regressed (most notably q72, which ran slower). Because Velox does not implement SortMergeJoin, it materialised millions of small columnar batches across downstream operators. In contrast, native Spark's whole-stage codegen fused the entire stream into a single JVM function without materialisation. This highlights that while Velox is incredibly fast, understanding your query patterns and join types is key to avoiding edge-case regressions.

The hardware acceleration benchmark: GPU-native Velox with NVIDIA cuDF:

By integrating NVIDIA's cuDF library directly into Velox, query plans from engines like Presto can be translated directly into GPU-accelerated pipelines.

- **Unmatched single-node performance:** On a TPC-H Scale Factor 1,000 dataset, a single node Presto C++ run on an AMD 7965WX CPU took 1,246 seconds. Shifting the entire Presto query plan to the GPU via Velox and cuDF reduced that runtime to 133.8 seconds on an NVIDIA RTX Pro 6000, and to a staggering 99.9 seconds on an NVIDIA GH200 Grace Hopper Superchip, representing an over 12x speedup.
- **Multi-GPU scaling with NVLink:** For distributed query execution, Velox supports a UCX based Exchange operator that runs the entire execution pipeline on the GPU. On an 8-GPU NVIDIA DGX A100 node, using high-bandwidth NVLink for data exchange yielded a >6x speedup compared to using Presto's baseline HTTP exchange.

Whether running on standard CPUs, modern Graviton processors, or state-of-the-art GPUs, Velox's vectorised C++ architecture delivers on its promises. By democratising elite

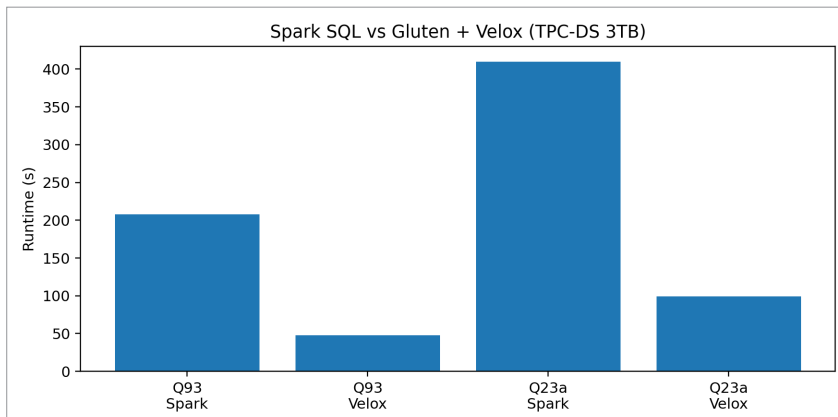


Figure 2: Distributed batch benchmark: Apache Spark + Gluten + Velox on AWS EKS

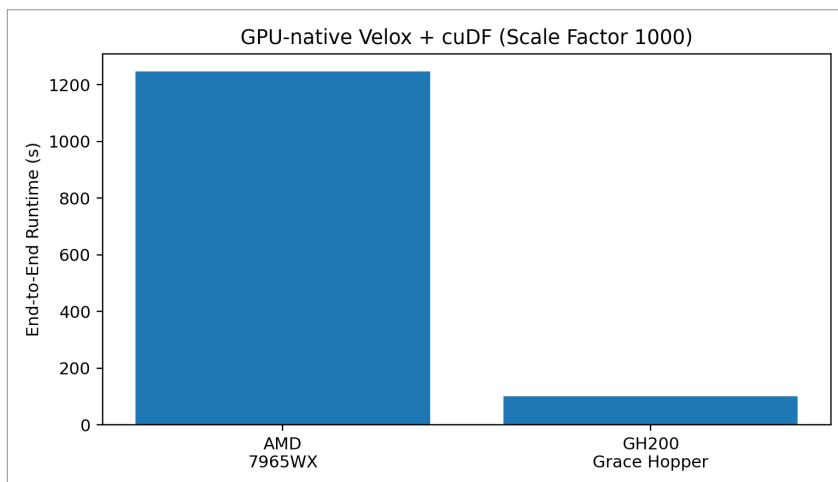


Figure 3: GPU-native Velox with NVIDIA cuDF